

Figure 3.43 Solenoid

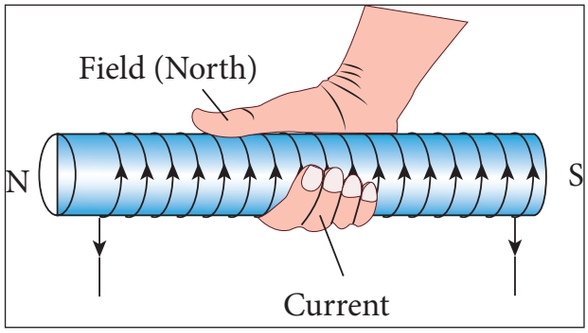


Figure 3.44 The direction of magnetic field of solenoid

solenoid looks like the magnetic field of a bar magnet.

The solenoid is assumed to be long which means that the length of the solenoid is large when compared to its diameter. The winding need not to be always circular, it can also be in other shapes. We consider here only circularly wound solenoid as shown in Figure 3.45.

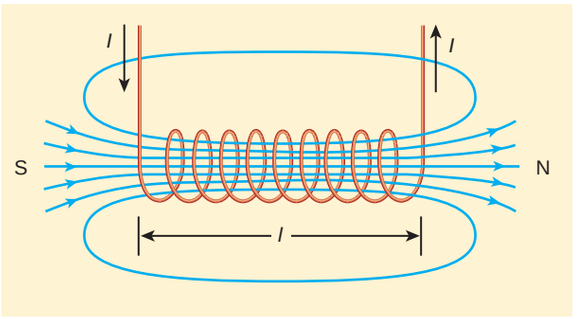


Figure 3.45 Solenoid as a bar magnet

3.9.3 Magnetic field due to a long current carrying solenoid

Consider a solenoid of length L having N turns. The diametre of the solenoid is assumed to be much smaller when compared to its length and the coil is wound very closely.

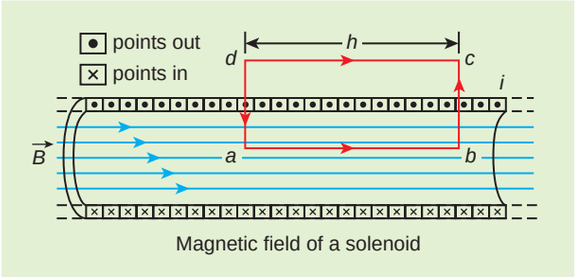


Figure 3.46 Amperian loop for solenoid

In order to calculate the magnetic field at any point inside the solenoid, we use Ampere's circuital law. Consider a rectangular loop abcd as shown in Figure 3.46. Then from Ampère's circuital law,

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